

## **Executive Summary**

This forensic investigation focused on analyzing suspicious network activity reported by ACME Incorporated involving internal host 192.168.10.4. The investigation was conducted using packet capture files and FTP server logs provided as evidence. Testing and analysis were performed without direct access or credentials to the victim system. During the investigation, multiple indicators of compromise and suspicious network activity were identified originating from internal host 192.168.10.5.

Analysis revealed evidence of ICMP host discovery, SYN port scanning, FTP interrogation, web enumeration, and persistent TCP communication with the victim system. The attacker identified the FTP service vsFTPD 2.3.4, a historically vulnerable FTP server version associated with known security vulnerabilities. FTP log analysis also identified multiple successful anonymous FTP login attempts from the attacker system. Although evidence of reconnaissance, scanning, and unauthorized access attempts was identified, no definitive evidence of successful data exfiltration was observed during the investigation.

## **Approach**

Between 05-06-2026 and 05-07-2026, Hashan Kodippilige performed a forensic investigation using Kali Linux and Wireshark to analyze FTP server logs and packet capture evidence provided by ACME Incorporated. The investigation was performed without direct access to the victim system and focused entirely on provided evidence files. Each packet capture and log entry was manually analyzed to identify suspicious behavior, attacker interactions, and possible indicators of compromise. The following methodology was used during the investigation.

1. Information Gathering
2. FTP Log Analysis
3. Network Traffic Analysis
4. Host Discovery Identification
5. Network Service Enumeration
6. Vulnerability Identification
7. Authentication and Access Analysis
8. Timeline Reconstruction
9. Attack Progression Analysis
10. Documentation and Reporting

## **Scope**

The scope of this investigation focused entirely on the evidence provided by ACME Incorporated, including packet capture files (file1.pcap through file5.pcap) and the FTP server log file client\_logs.log. The investigation included analysis of network traffic, FTP authentication activity, reconnaissance behavior, service scanning activity, and communication between the attacker and

victim systems. All analysis activities were performed within the authorized forensic investigation boundaries without modifying or interacting directly with production systems.

### **Assessment Overview & Recommendations**

The investigation revealed multiple indicators of compromise originating from internal host 192.168.10.5. The attacker performed ICMP host discovery, SYN port scanning, FTP interrogation, and HTTP web enumeration targeting the victim system 192.168.10.4. Multiple services including FTP, HTTP, HTTPS, SMB, and MySQL-related ports were targeted during scanning activity. The attacker successfully authenticated the FTP service using anonymous FTP access, which increased the risk of unauthorized access within the environment.

The FTP service running on the victim system was identified as vsFTPD 2.3.4, a historically vulnerable FTP server version associated with known security weaknesses. Additional packet captures showed persistent TCP communication and possible post-access interaction with the victim system. Although no definitive evidence of successful data exfiltration was identified, the observed activity strongly suggests unauthorized reconnaissance and possible compromise within the network environment.

Based on the findings of this investigation, the following recommendations should be implemented.

#### **Disable Anonymous FTP Access**

Anonymous FTP authentication should be disabled to prevent unauthorized access to internal systems.

#### **Patch or Remove Vulnerable Services**

Legacy and vulnerable services such as vsFTPD 2.3.4 should be patched, upgraded, or removed from the environment.

#### **Implement Stronger Authentication Policies**

Strong password policies and multi-factor authentication should be implemented where possible.

#### **Improve Internal Network Monitoring**

Internal traffic should be continuously monitored to detect suspicious scanning and reconnaissance activity.

#### **Deploy IDS/IPS Solutions**

Intrusion Detection and Prevention Systems should be deployed to improve visibility into malicious network activity.

#### **Segment Internal Networks**

Internal network segmentation should be implemented to reduce the risk of lateral movement between systems.

#### **Restrict Unnecessary Services and Ports**

Unused or unnecessary services and open ports should be disabled to reduce the attack surface of internal systems.